

Real-Time Sentiment Analytics over Multi-Room Chat

A WebSocket service combining lexicon sentiment and per-room aggregation under realistic load

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1. Abstract

Customer support, online community moderation, and team communication platforms all benefit from a low-latency sentiment signal layered over incoming messages. We describe and evaluate a WebSocket-based chat service that performs per-message sentiment analysis with VADER, computes per-room aggregates, and exposes them to clients with under 120 ms median end-to-end latency at 10,000 concurrent connections. Sentiment quality is evaluated against a 5,000-message labeled subset of the Cornell Movie Dialogs corpus annotated with crowdsourced polarity, achieving correlation 0.74 with the gold labels. Load tests with 100 simulated chat rooms and 100 messages per room per minute demonstrate stable p99 latency under 280 ms. The service reuses a single Socket.IO process with sticky-session affinity and an in-process LRU cache for room state.

Keywords: real-time analytics, sentiment, WebSockets, Socket.IO, lexicon-based

2. Introduction

Operators of high-volume chat platforms need a moderation and engagement signal that updates faster than batch sentiment models can deliver. Existing transformer-based services would be ideal for accuracy but introduce latency above the 200 ms threshold at which users perceive a system as responsive (Nielsen, 1993). The research problem is whether a lexicon-based sentiment scorer integrated into a Socket.IO chat service can deliver moderation-relevant signal at sub-200 ms p95 latency on commodity hardware while sustaining thousands of concurrent connections.

2.1 Research Problem

We additionally examine whether per-room sentiment aggregation produces an early warning signal for moderator intervention; this second question is the practical purpose of the system.

2.2 Research Questions and Hypotheses

Research question: Can the service hold p95 end-to-end latency below 200 ms at 10,000 concurrent connections?

Hypothesis: We hypothesize p95 latency under 180 ms based on benchmark profiles of Socket.IO at this scale, conditional on lexicon scoring being CPU-bound at sub-millisecond per-message cost.

Research question: Does the lexicon scorer correlate strongly enough with crowdsourced polarity labels to be useful for moderation?

Hypothesis: We expect Pearson correlation above 0.7 with crowd labels on the Cornell Movie Dialogs subset, sufficient for room-level moderation triggers.

Research question: Does per-room aggregate polarity provide a leading indicator of escalation events?

Hypothesis: We expect a negative shift in 5-minute rolling polarity to lead reported moderation flags by 60-180 seconds in the simulated dataset.

Research question: How does the service scale horizontally with sticky-session affinity versus a Redis-backed adapter?

Hypothesis: We expect sticky-session to scale linearly to ~50,000 connections per cluster and the Redis adapter to add ~25 ms per-hop latency for cross-process broadcast.

3. Literature Review

3.1 Theories Grounding the Problem

1. **Compositional Lexicon Sentiment (Hutto & Gilbert, 2014)** — VADER applies hand-curated rules over a polarity lexicon to handle negation, intensification, and contrastive conjunctions; this delivers sub-millisecond per-message scoring with explainable outputs. (Hutto & Gilbert (2014))
2. **Reactive Architectures (Bonér et al., 2014)** — Responsive, resilient, elastic, and message-driven systems form the foundation of modern real-time services; Socket.IO with non-blocking I/O is a canonical instantiation. (Bonér et al. (2014))
3. **Latency Perception Thresholds (Nielsen, 1993)** — Users perceive systems as instantaneous below 100 ms, responsive below 1 second, and broken above 10 seconds; chat applications must hold the 100-200 ms band end-to-end. (Nielsen (1993))
4. **Stochastic Modeling of Group Dynamics (Centola, 2010)** — Sentiment and behavior diffuse through chat networks; per-room polarity aggregates can serve as proxies for the underlying social dynamics in real time. (Centola (2010))
5. **Backpressure and Flow Control (Welsh et al., 2001)** — Staged event-driven architectures with explicit queues prevent cascading failure under burst load; this motivates the per-room ingest queue used in the implementation. (Welsh, Culler, & Brewer (2001))

3.2 Supporting Examples

- Discord publishes engineering posts on Socket.IO-style architectures handling millions of concurrent users; the design pattern translates downward to ten-thousand-user deployments.
- Twitch’s chat moderation platform integrates real-time sentiment for automated rule application, illustrating production demand for the artefact under study.
- Slack’s mention-tag features rely on similar low-latency event distribution; the same architecture composes naturally for sentiment notifications.

4. Research Method

The service is a Node.js Express application with Socket.IO for client connections. Each incoming message is scored by VADER in-process (no network hop). Per-room state is maintained in an in-memory Map with periodic JSON snapshotting for crash recovery. Load testing uses the Artillery WebSocket plugin: 100 simulated rooms, 100 users per room, 1 message every 10 seconds per user (10,000 concurrent connections, 1,000 messages per second peak). Sentiment quality is evaluated on a 5,000-message subset of the Cornell Movie Dialogs corpus with three-way crowdsourced polarity labels.

5. Data Description

Source: Cornell Movie Dialogs Corpus (annotated subset) plus simulated chat traffic — https://www.cs.cornell.edu/~cristian/Dialogs_Corpus.html

Coverage: Cornell sentiment subset: 5,000 utterances with 3-way (neg/neu/pos) crowd labels (3 annotators each); simulated traffic: 1.2 million messages over a 20-minute load test

Schema (selected fields):

- utterance_id, conversation_id, character_id, text
- crowd_polarity (mode of 3 annotators), polarity_agreement
- for simulated traffic: room_id, user_id, timestamp, message

Preprocessing: Cornell utterances were filtered to those with non-zero crowd agreement. Length was clipped to 280 characters to mirror typical chat-message sizes. Simulated traffic patterns were generated from an empirical inter-message time distribution fit on a public Discord public-server export.

License / availability: Cornell corpus: research-use license; simulated traffic synthesized.

6. Analysis

6.1 Lexicon sentiment quality vs crowd labels

Pearson correlation between VADER compound and crowd polarity on the 5,000-utterance evaluation subset.

Subset	n	Pearson r	MAE	3-way accuracy
All	5,000	0.74	0.21	0.78
High agreement (3/3)	2,840	0.81	0.16	0.87
Low agreement (2/3)	2,160	0.66	0.27	0.69

6.2 End-to-end latency under load

Latency is measured from message-publish to all-recipients-acknowledge across the swarm.

Concurrent users	p50 (ms)	p95 (ms)	p99 (ms)	CPU %
1,000	47	82	118	12
5,000	78	144	198	47
10,000	112	176	264	82
20,000 (Redis adapter)	138	221	312	61 per node $\times$ 4

6.3 Aggregate polarity as a leading indicator

We measured the time between a 5-minute rolling polarity drop of 0.3 absolute and a moderator-flagged escalation event in the simulated dataset.

Event class	n events	Mean lead time (s)	Hit rate	False alarm rate
Spam burst	84	112	0.92	0.07
Toxic argument	142	73	0.81	0.14
Off-topic drift	61	182	0.66	0.21

7. Discussion

The lexicon scorer is good enough at high crowd-agreement and adequate at low agreement; this is consistent with the difficulty of the underlying task (humans themselves disagree). End-to-end p95 latency stays under 200 ms through 10,000 concurrent users on a single node, and the Redis adapter scales horizontally with a modest latency penalty. Aggregate polarity is a valuable early warning signal for spam and toxic-argument escalations, less so for off-topic drift. The system is delivered with explicit moderator-alert thresholds tuned per event class on the simulated dataset.

8. Conclusion

A WebSocket service that integrates lexicon sentiment in-process and exposes per-room polarity aggregates can hold sub-200 ms p95 latency at 10,000 concurrent connections on a single node, with crowd-correlated sentiment quality of $r=0.74$. Aggregate polarity provides a usable early-warning signal for moderator intervention. The artefact is a sound default for moderate-scale chat moderation pipelines.

9. Future Work

- Replace VADER with a quantized DistilBERT scorer once GPU-backed inference cost falls below the latency budget.
- Add a per-user reputation prior to weight aggregate polarity by message author.
- Explore federated deployment so that room state can be sharded across nodes by hash(room_id).
- Extend the moderator-alert system with adaptive thresholds learned from feedback.

10. References

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